Contributions

- Further empirical evidence of heterogeneous price growth across space
  - Builds on e.g. Landvoigt, Piazessi, + Schneider
- Offers a new methodology for aggregating to the metro level
- Makes us think about units versus space aggregation
  - Highly salient to the "shoebox condo" debate

Target for Laspeyre's index

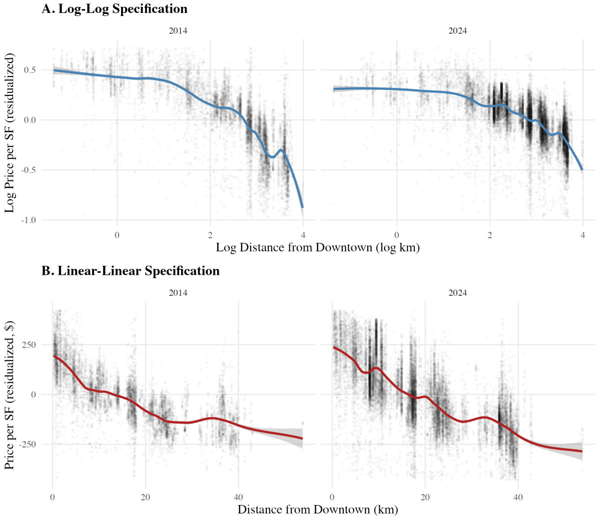
$$\frac{\Delta r_t^*}{r_{t-1}} = \frac{\sum_k r_t(k) h_{t-1}(k)/H_{t-1}(k)}{\sum_k r_{t-1}(k) h_{t-1}(k)/H_{t-1}(k)}.$$

- r price or rent, h individual home, H sum of housing units
- What is an h an important issue emphasized by authors
- Not easy to see common use where luxury homes get way more weight
- Observation: weights essentially an indicator for being transacted
  - More evidence on non-representativeness of that sample
    - Intuition for channel

Emphasis on location as heterogeneity

- Result from familiar monocentric city model
  - $\frac{dr(k)}{dk} = -\frac{\frac{d\text{commute cost}}{dk}}{h_t}$
- But linear commute cost posed unrealistic
  - 41st to 51st St in Manhattan vs random mile in the suburbs
- Suppose $\ln p(h,k) = h + \theta \log \frac{b}{k}$, then border growth yields identical price growth.
  - Joe Williams showed me this

Vancouver condo sales by year: distance gradients



Vancouver condo sales: (nonlinear) gradients change!

Dependent Variables:	log(ppsf)	ppsf
Model:	(1)	(2)
Variables		
Dist × year = 2011	-0.2327***	-8.635***
	(0.0013)	(0.0855)
Dist × year = 2012	-0.2390***	-8.790***
	(0.0011)	(0.0671)
Dist × year = 2013	-0.2397***	-8.747***
	(0.0008)	(0.0545)
Dist × year = 2014	-0.2444***	-9.529***
	(0.0008)	(0.0672)
Dist × year = 2015	-0.2518***	-9.933***
	(0.0008)	(0.0694)
Dist × year = 2016	-0.2816***	-12.38***
	(0.0010)	(0.0703)
Dist × year = 2017	-0.2784***	-14.07***
	(0.0012)	(0.0760)
Dist × year = 2018	-0.2410***	-13.48***
	(0.0012)	(0.0769)
Dist × year = 2019	-0.2382***	-13.29***
	(0.0011)	(0.0652)
Dist × year = 2020	-0.2327***	-13.05***
	(0.0011)	(0.0651)
Dist × year = 2021	-0.2132***	-12.88***
	(0.0010)	(0.0640)
Dist × year = 2022	-0.1847***	-11.45***
	(0.0027)	(0.1074)
Dist × year = 2023	-0.1817***	-11.89***
	(0.0038)	(0.1106)
Dist × year = 2024	-0.1693***	-10.92***
	(0.0027)	(0.0637)
Dist × year = 2025	-0.1721***	-11.58***
	(0.0015)	(0.0489)
Fixed-effects		
year	Yes	Yes
Fit statistics		
Observations	220,959	220,959
R^2	0.73021	0.74647
Within R^2	0.55930	0.58888

Clustered (year) standard errors in parentheses

Vancouver Condo: fit

- Log-log fits better 2014
- Subsequent years
  - Gradient flattens a lot
  - Action is at outer edge
  - Levels gets to be much better fit

Pairwise correlations among Vancouver vs suburban types

Type matters more than location?

	VanSingle	VanCondo	FVSingle	FVCondo
VanSingle	1.00	0.71	0.82	0.54
VanCondo	0.71	1.00	0.71	0.70
FVSingle	0.82	0.71	1.00	0.71
FVCondo	0.54	0.70	0.71	1.00

Useful reference? Nihilism vs hedonism?

Berliant/McMillen

- Might make same linear gradient choice?

So what would be helpful in intro/conclusion

Some ad hoc thougths

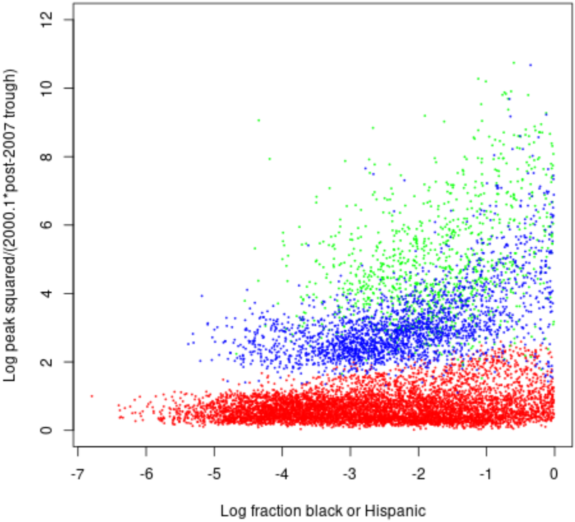
- What decision would be affected by different index choice?
- Portfolio efficiency for investor?
- Bailout decision for defaulters?
- Rationalize cap rate differences?

Maybe overly bold claim

- "insufficient attention has been paid to the implications of non-random sampling within cities"
- Many people were thinking about defaults polluting indexes (or lack thereof biasing upward) during GFC
- If no pubs maybe because no one found anything?
- Vague recollection I tried and found nothing
- But location of course within metro.
- See my review of Handbook of Urban Econ thing . . .

Zip code price movements boom bust (source, Zillow)

Red: Flyover, Blue: Coastal, green: Sand State



Price of services

"If the objective is to measure the price of housing services or housing space, then the definitions of hand H in the preceding equations should reflect a measure of housing services, rather than assuming that each housing unit provides the same quantity of housing space."

- Is the point that Bill Gates's house should get extra weight?

Common appreciation

- Standard monocentric gradient linear
  - Value at r = structure + lot size $\theta \int_{s=r}^{b} \frac{1}{\text{lot size at s}} ds$
  - Time log derivative unlikely constant by r
- But no less realistically (per Williams)
  $\log v = \log \text{structure} + \theta \left[\ln b(t) - \ln r \right]$
  - Then location not clear a source of growth heterogeneity
  - Might be fine with, say unit income elastcity
- Zoned single family vs multifamily clearer?

Proposition 2 doesn't really follow from Prop 1

- Are you missing "if not quasi-linearity, and if linear gradient in distance"?

Is AHS price data bananas?

- 2015-2019 Table 1 shows 18% vs 13% per year appreciation
- FHFA per FRED says about 7%
- Washington, D.C. Table 1 says about 2%
- FHFA per FRED says about 8%.
- Even I can afford Attom ...
- Aggregate Zillow zip indexes by metro central/suburbs?
- Given giant idiosyncratic errors, what can we rule out?
- Strong suggestion: use better data and see if differences as large